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JEE Main Physics Practice Sheet — DPP-053

Topic: Conservation of Angular Momentum & Its Applications

Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

Q1. A thin circular horizontal disc of mass M and radius R is rotating about its fixed central vertical axis with an angular speed ω_0 . Two small point masses, each of mass m , are placed gently at opposite ends of a diameter. The new angular speed of the system is:

- (A) $\frac{M}{M+2m}\omega_0$
- (B) $\frac{M}{M+m}\omega_0$
- (C) $\frac{M+4m}{M}\omega_0$
- (D) $\frac{M}{M+4m}\omega_0$

Q2. A person stands at the center of a circular platform rotating without friction about a vertical axis at 20 rpm with his arms outstretched holding heavy weights. When he pulls his arms inward, the moment of inertia of the system decreases to $\frac{2}{5}$ of its original value. The new rotational speed and the ratio of new to initial rotational kinetic energy are:

- (A) 50 rpm, 1.0
- (B) 50 rpm, 2.5
- (C) 8 rpm, 0.4
- (D) 25 rpm, 1.25

Q3. A horizontal turntable of mass M and radius R in the shape of a uniform disc rotates freely about a vertical frictionless axis through its center with angular velocity ω_0 . A child of mass m standing at the rim walks radially inward to the center. The angular velocity of the turntable when the child reaches the center is:

- (A) $\left(1 + \frac{m}{M}\right)\omega_0$
- (B) $\left(\frac{M}{M+2m}\right)\omega_0$
- (C) $\left(1 + \frac{2m}{M}\right)\omega_0$
- (D) $\left(\frac{M+m}{M}\right)\omega_0$

Q4. A uniform thin rod of mass M and length L lies on a smooth horizontal table and is free to rotate about a fixed vertical frictionless axis passing through one of its ends. A small particle of mass m moving horizontally with speed v_0 perpendicular to the rod strikes the free end and sticks to it. The angular velocity of the rod-particle system immediately after collision is:

- (A) $\frac{3mv_0}{(M+3m)L}$
- (B) $\frac{mv_0}{(M+m)L}$

(C) $\frac{6mv_0}{(M+3m)L}$

(D) $\frac{3mv_0}{(M+m)L}$

Q5. A uniform rod of mass M and length L is at rest on a frictionless horizontal plane (no fixed pivot). A particle of mass m moving horizontally with velocity v_0 perpendicular to the rod strikes one end of the rod and rebounds elastically with speed v_1 . The angular velocity ω of the rod about its center of mass just after the impact is:

- (A) $\frac{3m(v_0 + v_1)}{ML}$
- (B) $\frac{12m(v_0 + v_1)}{ML}$
- (C) $\frac{6m(v_0 - v_1)}{ML}$
- (D) $\frac{6m(v_0 + v_1)}{ML}$

Q6. A planet moves around the sun in an elliptical orbit. If r_p and r_a are the perihelion and aphelion distances and v_p and v_a are the corresponding orbital speeds, then by conservation of angular momentum:

- (A) $v_p r_a = v_a r_p$
- (B) $v_p^2 r_p = v_a^2 r_a$
- (C) $v_p r_p = v_a r_a$
- (D) $v_p \sqrt{r_p} = v_a \sqrt{r_a}$

Q7. A disc of moment of inertia I_1 is rotating with constant angular velocity ω_1 about a frictionless vertical axle. Another non-rotating disc of moment of inertia I_2 is dropped coaxially onto the first disc. The two discs eventually rotate together with common angular speed ω . The loss in kinetic energy during this process is:

- (A) $\frac{I_1 I_2}{2(I_1 + I_2)}\omega_1^2$
- (B) $\frac{(I_1 + I_2)}{2I_1 I_2}\omega_1^2$
- (C) $\frac{I_1^2}{2(I_1 + I_2)}\omega_1^2$
- (D) $\frac{I_2^2}{2(I_1 + I_2)}\omega_1^2$

Q8. A small mass m attached to a string passes through a small hole in a frictionless horizontal table. Initially, the mass revolves in a circle of radius r_1 with speed v_1 . The string is slowly pulled from below so that the radius decreases to $r_2 = \frac{r_1}{2}$. The new speed of the mass and the work done on the system are:

- (A) $4v_1, \frac{15}{2}mv_1^2$

- (B) $2v_1, \frac{3}{2}mv_1^2$
 (C) $2v_1, \frac{1}{2}mv_1^2$
 (D) $v_1, 0$

Q9. A circular platform is free to rotate about a frictionless vertical axis. A man walks around the perimeter of the platform with a velocity v relative to the platform. If the moment of inertia of the platform is I , its radius is R , and the mass of the man is m , the absolute angular velocity of the platform (starting from rest) is:

- (A) $\frac{mvR}{I}$
 (B) $\frac{mvR}{I - mR^2}$
 (C) $\frac{mv}{R(I + mR^2)}$
 (D) $\frac{mvR}{I + mR^2}$

Q10. A uniform solid sphere of mass M and radius R rotating with initial angular velocity ω_0 about a horizontal axis through its center is gently placed on a rough horizontal floor (μ). It initially slips and eventually rolls without slipping. The angular velocity of the sphere when pure rolling sets in is:

- (A) $\frac{5}{7}\omega_0$
 (B) $\frac{2}{5}\omega_0$
 (C) $\frac{2}{7}\omega_0$
 (D) $\frac{3}{7}\omega_0$

Q11. A particle of mass m is projected with speed u at an angle θ with the horizontal. The magnitude of the angular momentum of the projectile about the point of projection when the particle reaches its maximum height is:

- (A) $\frac{mu^3 \sin^2 \theta \cos \theta}{2g}$
 (B) $\frac{mu^3 \sin \theta \cos^2 \theta}{2g}$
 (C) $\frac{mu^3 \sin^2 \theta \cos \theta}{g}$
 (D) $\frac{mu^3 \cos^3 \theta}{2g}$

Q12. A uniform thin square plate of mass M and side a rotates in a horizontal plane with angular speed ω about a vertical axis passing through its center. Four identical point masses m are placed at the four vertices of the square. The new angular speed is:

- (A) $\frac{M}{M + 6m}\omega$
 (B) $\frac{M}{M + 12m}\omega$
 (C) $\frac{M}{M + 3m}\omega$

(D) $\frac{M}{M + 4m}\omega$

Q13. A particle of mass m moves in a central force field where the potential energy is $U(r) = kr^2$. Which of the following quantities is strictly conserved throughout its motion?

- (A) Linear momentum $\vec{p}$
 (B) Kinetic energy K
 (C) Velocity vector $\vec{v}$
 (D) Angular momentum $\vec{L}$ about the center of force

Q14. A uniform rod of mass M and length L is pivoted at its center so that it can rotate freely in a horizontal plane. Initially at rest, two small lumps of clay of mass m each travelling with speed v in opposite directions hit the two ends simultaneously and stick to them. The resulting angular speed is:

- (A) $\frac{6mv}{(M + 6m)L}$
 (B) $\frac{6mv}{(M + 12m)L}$
 (C) $\frac{12mv}{(M + 6m)L}$
 (D) $\frac{3mv}{(M + 6m)L}$

Q15. A hollow spherical shell of mass M and radius R is spinning with angular velocity ω_0 about a diameter. Suddenly, its radius shrinks uniformly to $\frac{R}{2}$ while keeping its mass constant. The new rotational kinetic energy of the shell is:

- (A) 2 times the initial value
 (B) 8 times the initial value
 (C) Equal to the initial value
 (D) 4 times the initial value

Q16. A uniform disc of mass M and radius R is rolling without slipping on a horizontal plane with translation speed v . The angular momentum of the disc about the instantaneous point of contact with the ground is:

- (A) $\frac{3}{2}MvR$
 (B) $\frac{1}{2}MvR$
 (C) MvR
 (D) $2MvR$

Q17. A gymnast on a high bar begins rotating with an angular velocity ω_1 while in an extended posture. When she tucks her body into a compact ball, her moment of inertia reduces by a factor of 3. Neglecting air resistance and friction at the hands:

- (A) Angular momentum triples and kinetic energy remains constant
 (B) Angular velocity triples and kinetic energy triples
 (C) Angular velocity triples and kinetic energy decreases to one-third

- (D) Angular velocity and kinetic energy remain constant
- Q18.** A uniform cylinder of radius R is spun about its longitudinal symmetry axis to an angular speed ω_0 and then placed gently into a corner between a horizontal floor and a vertical wall with identical coefficient of friction μ . If the net external torque about the center of the cylinder is non-zero:
- (A) Angular momentum is conserved about the center
 - (B) Angular momentum is conserved about the contact point on the floor
 - (C) Angular momentum decreases to zero as mechanical energy dissipates as heat
 - (D) Angular velocity remains ω_0
- Q19.** A circular hoop of mass M and radius R is rotating about its central vertical axis with angular speed ω_0 .
- An identical stationary hoop is gently dropped coaxially onto it. If friction causes them to rotate together, the fraction of initial rotational kinetic energy lost is:
- (A) 25%
 - (B) 75%
 - (C) 0%
 - (D) 50%
- Q20.** A particle of mass m moves with constant velocity $\vec{v} = v_0\hat{i}$ along the straight line $y = d, z = 0$. The angular momentum of the particle about the origin $\vec{L}_O$ as a function of time t is:
- (A) Constant and equal to $+mv_0d\hat{k}$
 - (B) Decreases inversely with distance from origin
 - (C) Constant and equal to $-mv_0d\hat{k}$
 - (D) Increases linearly with time as $mv_0(v_0t)\hat{k}$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-053)

Conservation of Angular Momentum & Its Applications

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	D	6	C	11	A	16	A
2	B	7	A	12	B	17	B
3	C	8	B	13	D	18	C
4	A	9	D	14	C	19	D
5	D	10	C	15	D	20	C

Detailed Step-by-Step Solutions

Sol 1. Option (D)

Initial moment of inertia: $I_i = \frac{1}{2}MR^2$.

When two masses m are placed at opposite ends of a diameter (distance R from axis):

$$I_f = \frac{1}{2}MR^2 + 2(mR^2) = \left(\frac{M+4m}{2}\right)R^2$$

Since no external torque acts about the vertical axis ($\tau_{\text{ext}} = 0$):

$$I_i\omega_0 = I_f\omega \Rightarrow \left(\frac{1}{2}MR^2\right)\omega_0 = \left(\frac{M+4m}{2}\right)R^2\omega$$

$$\omega = \frac{M}{M+4m}\omega_0$$

Sol 2. Option (B)

Given $I_f = \frac{2}{5}I_i$. By conservation of angular momentum:

$$I_i\omega_i = I_f\omega_f \Rightarrow \omega_f = \left(\frac{I_i}{I_f}\right)\omega_i = \left(\frac{5}{2}\right)(20 \text{ rpm}) = 50 \text{ rpm}$$

Ratio of rotational kinetic energy:

$$\frac{K_f}{K_i} = \frac{\frac{1}{2}I_f\omega_f^2}{\frac{1}{2}I_i\omega_i^2} = \frac{I_f}{I_i} = \frac{5}{2} = 2.5$$

The kinetic energy increases because internal muscular work is done by the person pulling his arms inward.

Sol 3. Option (C)

Initial moment of inertia with child at rim (distance R):

$$I_i = \frac{1}{2}MR^2 + mR^2 = \left(\frac{M+2m}{2}\right)R^2$$

Final moment of inertia when child is at the center ($r = 0$):

$$I_f = \frac{1}{2}MR^2$$

By conservation of angular momentum:

$$I_i\omega_0 = I_f\omega \Rightarrow \left(\frac{M+2m}{2}\right)R^2\omega_0 = \left(\frac{M}{2}\right)R^2\omega$$

$$\omega = \left(\frac{M+2m}{M}\right)\omega_0 = \left(1 + \frac{2m}{M}\right)\omega_0$$

Sol 4. Option (A)

About the fixed pivot at the end, external torque during

impact is zero.

Initial angular momentum about pivot:

$$L_i = mv_0L$$

Total moment of inertia of rod and embedded mass about pivot:

$$I_{\text{total}} = I_{\text{rod}} + I_{\text{particle}} = \frac{1}{3}ML^2 + mL^2 = \left(\frac{M+3m}{3}\right)L^2$$

By conservation of angular momentum about pivot:

$$mv_0L = \left(\frac{M+3m}{3}\right)L^2\omega \Rightarrow \omega = \frac{3mv_0}{(M+3m)L}$$

Sol 5. Option (D)

Since no external force/torque acts on the free rod, angular momentum about the center of mass (CM) of the rod is conserved.

Initial angular momentum of the system about CM: $L_i = mv_0(L/2)$.

Final angular momentum of particle about CM: $L_{p,f} = -mv_1(L/2)$ (opposite direction).

Final angular momentum of rod about CM: $L_{\text{rod}} = I_{\text{cm}}\omega = \left(\frac{1}{12}ML^2\right)\omega$.

Conservation of angular momentum about CM:

$$mv_0\frac{L}{2} = -mv_1\frac{L}{2} + \frac{1}{12}ML^2\omega$$

$$\frac{1}{12}ML^2\omega = \frac{m(v_0 + v_1)L}{2} \Rightarrow \omega = \frac{6m(v_0 + v_1)}{ML}$$

Sol 6. Option (C)

Gravitational force by the Sun is purely central ($\vec{\tau} = \vec{r} \times \vec{F} = \vec{0}$), so the planet's angular momentum about the Sun is strictly conserved:

$$L = mv_p r_p = mv_a r_a \Rightarrow v_p r_p = v_a r_a$$

Sol 7. Option (A)

By conservation of angular momentum about the rotation axis:

$$I_1\omega_1 = (I_1 + I_2)\omega \Rightarrow \omega = \frac{I_1\omega_1}{I_1 + I_2}$$

Initial kinetic energy: $K_i = \frac{1}{2}I_1\omega_1^2$.

Final kinetic energy:

$$K_f = \frac{1}{2}(I_1 + I_2)\omega^2 = \frac{1}{2}(I_1 + I_2)\left(\frac{I_1\omega_1}{I_1 + I_2}\right)^2 = \frac{1}{2}\frac{I_1^2\omega_1^2}{I_1 + I_2}$$

Loss in kinetic energy $\Delta K = K_i - K_f$:

$$\Delta K = \frac{1}{2} I_1 \omega_1^2 \left(1 - \frac{I_1}{I_1 + I_2} \right) = \frac{I_1 I_2}{2(I_1 + I_2)} \omega_1^2$$

Sol 8. Option (B)

Tension in the string acts radially through the hole, so torque about the central hole is zero ($\tau = 0$).

Conservation of angular momentum:

$$mv_1 r_1 = mv_2 r_2 \implies v_2 = v_1 \left(\frac{r_1}{r_2} \right) = v_1(2) = 2v_1$$

By Work-Energy Theorem, work done by pulling the string:

$$W = \Delta K = \frac{1}{2} mv_2^2 - \frac{1}{2} mv_1^2 = \frac{1}{2} m(4v_1^2 - v_1^2) = \frac{3}{2} mv_1^2$$

Sol 9. Option (D)

Let Ω be the angular velocity of the platform relative to ground in the opposite direction.

The absolute linear velocity of the man relative to ground is $v_{\text{man}} = v - \Omega R$.

Total angular momentum of the isolated system about the vertical axis is conserved ($L_i = 0$):

$$0 = I\Omega - mv_{\text{man}}R \implies I\Omega = m(v - \Omega R)R$$

$$(I + mR^2)\Omega = mvR \implies \Omega = \frac{mvR}{I + mR^2}$$

Sol 10. Option (C)

The friction force f_k and normal reaction N pass through the contact line on the floor. Therefore, the torque of all external forces about any point on the contact line on the floor is zero.

Initial angular momentum about the contact point on floor:

$$L_i = I_{\text{cm}}\omega_0 + mv_0R = \left(\frac{2}{5}MR^2 \right) \omega_0 + 0 = \frac{2}{5}MR^2\omega_0$$

When pure rolling starts ($v = \omega R$):

$$L_f = I_{\text{contact}}\omega = \left(\frac{2}{5}MR^2 + MR^2 \right) \omega = \frac{7}{5}MR^2\omega$$

Equating $L_i = L_f$:

$$\frac{2}{5}MR^2\omega_0 = \frac{7}{5}MR^2\omega \implies \omega = \frac{2}{7}\omega_0$$

Sol 11. Option (A)

At maximum height, velocity is purely horizontal: $v_x = u \cos \theta$.

Maximum vertical height: $H = \frac{u^2 \sin^2 \theta}{2g}$.

The perpendicular distance from the projection point to the line of velocity at the highest point is H .

Angular momentum about the origin:

$$L = mv_x H = m(u \cos \theta) \left(\frac{u^2 \sin^2 \theta}{2g} \right) = \frac{mu^3 \sin^2 \theta \cos \theta}{2g}$$

Sol 12. Option (B)

Moment of inertia of square plate about central perpendicular axis: $I_{\text{plate}} = \frac{1}{6}Ma^2$.

Distance of each vertex from center: $r = \frac{a}{\sqrt{2}} \implies r^2 = \frac{a^2}{2}$.

Moment of inertia of 4 point masses: $I_{\text{masses}} = 4 \times mr^2 = 4m \left(\frac{a^2}{2} \right) = 2ma^2$.

Total final moment of inertia:

$$I_f = \frac{1}{6}Ma^2 + 2ma^2 = \left(\frac{M + 12m}{6} \right) a^2$$

Conservation of angular momentum:

$$\left(\frac{1}{6}Ma^2 \right) \omega = \left(\frac{M + 12m}{6} \right) a^2 \omega' \implies \omega' = \frac{M}{M + 12m} \omega$$

Sol 13. Option (D)

For any central force field $\vec{F}(\vec{r}) = f(r)\hat{r}$, the torque about the center of force is identically zero: $\vec{\tau} = \vec{r} \times \vec{F} = \vec{0}$.

Hence, the angular momentum $\vec{L}$ about the center of force is strictly conserved.

Sol 14. Option (C)

Initial angular momentum about the central pivot:

$$L_i = 2 \times mv \left(\frac{L}{2} \right) = mvL$$

Total moment of inertia about the central pivot after adhesion:

$$I_f = \frac{1}{12}ML^2 + 2m \left(\frac{L}{2} \right)^2 = \left(\frac{M + 6m}{12} \right) L^2$$

Conservation of angular momentum:

$$mvL = \left(\frac{M + 6m}{12} \right) L^2 \omega \implies \omega = \frac{12mv}{(M + 6m)L}$$

Sol 15. Option (D)

Moment of inertia of spherical shell: $I = \frac{2}{3}MR^2$.

When radius becomes $R' = R/2$: $I_f = \frac{2}{3}M(R/2)^2 = \frac{I_i}{4}$.

By angular momentum conservation:

$$I_i \omega_0 = I_f \omega_f \implies \omega_f = 4\omega_0$$

New rotational kinetic energy:

$$K_f = \frac{1}{2} I_f \omega_f^2 = \frac{1}{2} \left(\frac{I_i}{4} \right) (4\omega_0)^2 = 4 \left(\frac{1}{2} I_i \omega_0^2 \right) = 4K_i$$

Sol 16. Option (A)

Using the parallel-axis theorem about the instantaneous contact point P :

$$I_P = I_{\text{cm}} + MR^2 = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$$

Since rolling without slipping means $\omega = \frac{v}{R}$:

$$L_P = I_P \omega = \left(\frac{3}{2}MR^2 \right) \left(\frac{v}{R} \right) = \frac{3}{2}MvR$$

Sol 17. Option (B)

No external torque acts on the gymnast ($L = I\omega = \text{constant}$).

Since $I_2 = \frac{I_1}{3}$, angular velocity becomes $\omega_2 = 3\omega_1$ (triples). Rotational kinetic energy:

$$K_2 = \frac{L^2}{2I_2} = \frac{L^2}{2(I_1/3)} = 3 \left(\frac{L^2}{2I_1} \right) = 3K_1$$

Both angular velocity and rotational kinetic energy triple.

Sol 18. Option (C)

Frictional forces from the wall and floor create an opposing external retarding torque about the axis of rotation.

Since $\tau_{\text{ext}} \neq 0$, angular momentum is not conserved and continuously decreases until the cylinder comes to rest, with initial kinetic energy converted to thermal energy.

Sol 19. Option (D)

For a thin hoop: $I_1 = I_2 = MR^2$.

By conservation of angular momentum:

$$I_1\omega_0 = (I_1 + I_2)\omega_f \implies (MR^2)\omega_0 = (2MR^2)\omega_f \implies \omega_f = \frac{\omega_0}{2}$$

Final kinetic energy:

$$K_f = \frac{1}{2}(2MR^2)\left(\frac{\omega_0}{2}\right)^2 = \frac{1}{4}MR^2\omega_0^2 = \frac{1}{2}K_i$$

Fraction of kinetic energy lost is $1 - 0.5 = 50\%$.

Sol 20. Option (C)

Position vector of particle: $\vec{r}(t) = (v_0t)\hat{i} + d\hat{j}$.

Linear momentum: $\vec{p} = mv_0\hat{i}$.

Angular momentum about origin:

$$\vec{L}_O = \vec{r} \times \vec{p} = \left((v_0t)\hat{i} + d\hat{j}\right) \times (mv_0\hat{i}) = -mv_0d\hat{k}$$

The angular momentum is completely independent of time and remains constant at $-mv_0d\hat{k}$.